Group - II crossection derivation (corrected)

Corrections (blue; checked against Calculation_main.tex, see Groups_II_III_check.pdf):

- Row I (parts 1 and 2) and Row IV up to the z integration: correct. Only the integration measures are completed.
- Row IV, z integration and fragmentation form: replaced by the exact result (no limits on $\vee, k^+, \Lambda$).
- Not changed here: index placements of the adjoint Wilson lines. They are correct with your convention $U^{bc} = U^{cb}$; for a general adjoint U see Sec. 5 of the check note.

1 Group II

1.1 Row I [Done ✓]

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{4}{(2\pi)^3}\int_\Lambda^{\vee - k^+}\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\left[\mathbb{B}^{\dagger\,bc'}_{2ji}(k^+,-w';p^+,-z)U^{c'a}(w') - \bar{\mathbb{B}}^{\dagger\,c'a}_{2ji}(k^+,-w';p^+,-z)U^{bc'}(z)\right]\times$$
$$\left[U^{ac}(w)\mathbb{B}^{bc}_{2ji}(k^+,-w;p^+,-z) - U^{\dagger\,bc}(z)\bar{\mathbb{B}}^{ca}_{2ji}(k^+,-w;p^+,-z)\right]$$

We write it in two parts:

1.1.1 First part: [Done ✓]

We write it in two parts:

$$\frac{d^3N_{1\,\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{ig^4}{8\pi^4}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\frac{1}{[p^+(x-z)^2+k^+(x-w)^2]}$$
$$\left[\frac{\delta^{ij}\left[(x-z)^2-(x-w)^2\right]}{2\,(p^++k^+)\,(z-w)^2}+\frac{1}{p^+}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2}\right]+\frac{1}{k^+}\left[\frac{(x-z)^j\,(z-w)^i}{(z-w)^2}\right.\right.$$
$$\left.\left.-\frac{(x-z)^j\,(x-w)^i}{2\,(x-w)^2}\right]\right]\times\left[U^{ac'}(w')\,\{\rho^{c'}(x'),\rho^b(y)\}-U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\right]\times$$
$$\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]$$
$$+\frac{1}{(2\pi)^3}\frac{ig^4}{8\pi^4}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\frac{1}{[p^+(x-z)^2+k^+(x-w')^2]}$$
$$\left[\frac{\delta^{ij}\left[(x-z)^2-(x-w')^2\right]}{2\,(p^++k^+)\,(z-w')^2}+\frac{1}{p^+}\left[\frac{(x-w')^i\,(z-w')^j}{(z-w')^2}+\frac{(x-z)^j\,(x-w')^i}{2\,(x-z)^2}\right]+\frac{1}{k^+}\left[\frac{(x-z)^j\,(z-w')^i}{(z-w')^2}\right.\right.$$
$$\left.\left.-\frac{(x-z)^j\,(x-w')^i}{2\,(x-w')^2}\right]\right]\times\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x)-f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,\rho^{e'}(x)\right]\times$$
$$\left[U^{ac}(w)\,\{\rho^c(x'),\rho^b(y)\}-U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\{\rho^d(x'),\rho^e(y)\}\right] \tag{1.1}$$

Integrating w.r.t. p^+

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\Bigg\{ & \\
-\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\log\left[\frac{(\vee-k^+)\,(x-z)^2+k^+\,(x-w)^2}{\Lambda\,(x-z)^2+k^+\,(x-w)^2}\right]\times & \\
\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2}-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right] & \\
+\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda+k^+}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2} & \\
-\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[1-\frac{k^+}{\vee}\right]\right]\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\} & \\
\left[U^{ac'}(w')\,\{\rho^{c'}(x'),\rho^b(y)\}-U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right] & \\
+\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\Bigg\{ & \\
\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\,\log\left[\frac{(\vee-k^+)\,(x-z)^2+k^+\,(x-w')^2}{\Lambda\,(x-z)^2+k^+\,(x-w')^2}\right]\times & \\
\left[\frac{\delta^{ij}}{2\,(z-w')^2}+\frac{(x-z)^j\,(z-w')^i}{(z-w')^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w')^i}{(x-w')^2\,(x-z)^2}-\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}\right] & \\
-\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda+k^+}\right]\frac{(x'-w)\cdot(y-z)}{(x'-w)^2\,(y-z)^2\,(z-w')^2} & \\
+\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[1-\frac{k^+}{\vee}\right]\right]\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\,\left[\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}+\frac{(x-z)^j\,(x-w')^i}{2\,(x-z)^2\,(x-w')^2}\right]\Bigg\} & \\
\times\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x)-f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,\rho^{e'}(x)\right]\times\left[U^{ac}(w)\,\{\rho^c(x'),\rho^b(y)\}-U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\{\rho^d(x'),\rho^e(y)\}\right] &
\end{aligned}
\tag{1.2}
$$

1.1.2 Second part: [Done ✓]

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\frac{p^+\,k^+}{(p^++k^+)^2}\,\frac{1}{[p^+\,(x-z)^2+k^+\,(x-w)^2]}\,\frac{1}{[p^+\,(x'-z)^2+k^+\,(x'-w')^2]} & \\
\Bigg\{\left[\frac{\delta^{ij}\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2}+\frac{(p^++k^+)}{k^+}\,\left[\frac{(x-z)^j\,(z-w)^i}{(z-w)^2}-\frac{(x-z)^j\,(x-w)^i}{2\,(x-w)^2}\right]+\frac{(p^++k^+)}{p^+}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2}\right.\right. & \\
\left.\left.+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2}\right]\right]\times\left[\frac{\delta^{ij}\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w')^2}+\frac{(p^++k^+)}{k^+}\,\left[\frac{(x'-z)^j\,(z-w')^i}{(z-w')^2}-\frac{(x'-z)^j\,(x'-w')^i}{2\,(x'-w')^2}\right]\right. & \\
\left.+\frac{(p^++k^+)}{p^+}\,\left[\frac{(x'-w')^i\,(z-w')^j}{(z-w')^2}+\frac{(x'-z)^j\,(x'-w')^i}{2\,(x'-z)^2}\right]\right]\Bigg\}\times\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x')-f^{ac'd'}\,U^{bc'}(z)U^{e'd'}(x')\,\rho^{e'}(x')\right] & \\
\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right] &
\end{aligned}
\tag{1.3}
$$

Reducing the Wilson lines,

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\frac{p^+\,k^+}{(p^++k^+)^2}\,\frac{1}{[p^+\,(x-z)^2+k^+\,(x-w)^2]}\,\frac{1}{[p^+\,(x'-z)^2+k^+\,(x'-w')^2]}$$
$$\left\{\left[\frac{\delta^{ij}\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2}+\frac{(p^++k^+)}{k^+}\,\left[\frac{(x-z)^j\,(z-w)^i}{(z-w)^2}-\frac{(x-z)^j\,(x-w)^i}{2\,(x-w)^2}\right]+\frac{(p^++k^+)}{p^+}\,\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2}\right.\right.\right.$$
$$\left.\left.\left.+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2}\right]\right]\times\left[\frac{\delta^{ij}\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w')^2}+\frac{(p^++k^+)}{k^+}\,\left[\frac{(x'-z)^j\,(z-w')^i}{(z-w')^2}-\frac{(x'-z)^j\,(x'-w')^i}{2\,(x'-w')^2}\right]\right.$$
$$\left.+\frac{(p^++k^+)}{p^+}\,\left[\frac{(x'-w')^i\,(z-w')^j}{(z-w')^2}+\frac{(x'-z)^j\,(x'-w')^i}{2\,(x'-z)^2}\right]\right]\right\}\times\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^d(x)\right.$$
$$-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^e(x)-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^d(x)$$
$$\left.+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^e(x)\right] \tag{1.4}$$

Simplifying, we get

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\frac{p^+\,k^+}{(p^++k^+)^2}\,\frac{1}{[p^+\,(x-z)^2+k^+\,(x-w)^2]}\,\frac{1}{[p^+\,(x'-z)^2+k^+\,(x'-w')^2]}$$
$$\left\{\frac{d_\perp\,\left[(x-z)^2-(x-w)^2\right]\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(z-w)^2\,(z-w')^2}\right.$$
$$+\frac{(p^++k^+)}{k^+}\left[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\right.$$
$$\left.+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\right]$$
$$+\frac{(p^++k^+)}{p^+}\left[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\right.$$
$$\left.+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\right]$$
$$+\frac{(p^++k^+)^2}{k^{+\,2}}\left[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2}\right.$$
$$\left.-\frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\right]$$
$$+\frac{(p^++k^+)^2}{p^{+\,2}}\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.$$
$$\left.+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]$$
$$+\frac{(p^++k^+)^2}{p^+k^+}\left[\frac{(x-w)\cdot(z-w')\,(x'-z)\cdot(z-w)}{(z-w)^2\,(z-w')^2}-\frac{(x-w)\cdot(x'-w')\,(z-w)\cdot(x'-z)}{2\,(z-w)^2\,(x'-w')^2}\right.$$
$$+\frac{(x-z)\cdot(x'-z)\,(x-w)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-w')^2}$$
$$+\frac{(x'-w')\cdot(z-w)\,(x-z)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(z-w)\,(x-z)\cdot(x'-z)}{2\,(z-w)^2\,(x'-z)^2}$$
$$\left.\left.-\frac{(x-z)\cdot(z-w')\,(x'-w')\cdot(x-w)}{2\,(x-w)^2\,(z-w')^2}-\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(x-z)}{4\,(x-w)^2\,(x'-z)^2}\right]\right\}\times$$
$$\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^d(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^e(x)\right.$$
$$\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^d(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^e(x)\right] \tag{1.5}$$

Integrating w.r.t. p^+ and we write it in 6 different parts:

$$\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\left\{\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\times\right.$$
$$\left[\frac{(x-w)\cdot(z-w')\,(x'-z)\cdot(z-w)}{(z-w)^2\,(z-w')^2}-\frac{(x-w)\cdot(x'-w')\,(z-w)\cdot(x'-z)}{2\,(z-w)^2\,(x'-w')^2}\right.$$
$$+\frac{(x-z)\cdot(x'-z)\,(x-w)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-w')^2}$$
$$+\frac{(x'-w')\cdot(z-w)\,(x-z)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(z-w)\,(x-z)\cdot(x'-z)}{2\,(z-w)^2\,(x'-z)^2}$$
$$\left.-\frac{(x-z)\cdot(z-w')\,(x'-w')\cdot(x-w)}{2\,(x-w)^2\,(z-w')^2}-\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(x-z)}{4\,(x-w)^2\,(x'-z)^2}\right]\times$$
$$\left.\left[\log\left[\frac{(\vee-k^+)\,(x-z)^2+k^+\,(x-w)^2}{k^+\,(x'-w')^2+(\vee-k^+)\,(x'-z)^2}\right]-\log\left[\frac{\Lambda\,(x-z)^2+k^+\,(x-w)^2}{k^+\,(x'-w')^2+\Lambda\,(x'-z)^2}\right]\right]\right\}\times$$
$$\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.$$
$$\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \tag{1.6}$$

$$\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[1-\frac{k^+}{\vee}\right]\right]\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{1}{[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2]}\times$$
$$\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.$$
$$\left.+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\left[(x-z)^2-\frac{(x-w)^2\,(x'-z)^2}{(x'-w')^2}\right]\times$$
$$\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.$$
$$\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]$$
$$-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-z)^2}{[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2]}\times$$
$$\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.$$
$$\left.+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\log\left[\frac{(\vee-k^+)\,(x-z)^2+k^+\,(x-w)^2}{\Lambda\,(x-z)^2+k^+\,(x-w)^2}\right]\times$$
$$\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.$$
$$\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]$$
$$+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-z)^2}{[(x'-w')^4\,(x-z)^2\,-(x-w)^2\,(x'-z)^2\,(x'-w')^2]}\times$$
$$\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.$$
$$\left.+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\log\left[\frac{(\vee-k^+)\,(x'-z)^2+k^+\,(x'-w')^2}{\Lambda\,(x'-z)^2+k^+\,(x'-w')^2}\right]$$
$$\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.$$
$$\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \tag{1.7}$$

$$
\begin{aligned}
\frac{d^3N_{2c\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')^2}{(x'-z)^2}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]} \\
&\left[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2}-\frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}\right. \\
&\left.+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\right]\times\log\left[\frac{(\vee-k^+)\,(x'-z)^2+k^+\,(x'-w')^2}{\Lambda\,(x'-z)^2+k^+\,(x'-w')^2}\right] \\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right. \\
&\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-w)^2}{(x-z)^2}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]} \\
&\left[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2}-\frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}\right. \\
&\left.+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\right]\times\log\left[\frac{(\vee-k^+)\,(x-z)^2+k^+\,(x-w)^2}{\Lambda\,(x-z)^2+k^+\,(x-w)^2}\right] \\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right. \\
&\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \qquad (1.8)
\end{aligned}
$$

$$
\begin{aligned}
\frac{d^3N_{2d\,\mathrm{NLO}}}{d^3k} = &-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+ + \Lambda}\right]\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{1}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2-(x'-z)^2]} \\
&\left[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\right. \\
&\left. + \frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\right] \\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right. \\
&\left. - f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x) + N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \\
&+ \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-z)^2}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]} \\
&\left[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\right. \\
&\left. + \frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\right]\times \\
&\log\left[\frac{(\vee-k^+)\,(x-z)^2+k^+\,(x-w)^2}{\Lambda\,(x-z)^2+k^+\,(x-w)^2}\right]\times \\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right. \\
&\left. - f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x) + N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \\
&+ \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-z)^2}{[(x'-w')^2-(x'-z)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]} \\
&\left[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\right. \\
&\left. + \frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\right]\times \\
&\log\left[\frac{(\vee-k^+)\,(x'-z)^2+k^+\,(x'-w')^2}{\Lambda\,(x'-z)^2+k^+\,(x'-w')^2}\right]\times \\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right. \\
&\left. - f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x) + N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \qquad (1.9)
\end{aligned}
$$

$$
\begin{aligned}
\frac{d^3N_{2e\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda+k^+}\right]\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{1}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2-(x'-z)^2]}\\
&\left[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\right.\\
&\left.+\,\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\right]\\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.\\
&\left.-\,f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-w)^2}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\\
&\left[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\right.\\
&\left.+\,\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\right]\times\\
&\log\left[\frac{(\vee-k^+)\,(x-z)^2+k^+\,(x-w)^2}{\Lambda\,(x-z)^2+k^+\,(x-w)^2}\right]\times\\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.\\
&\left.-\,f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')^2}{[(x'-w')^2-(x'-z)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\\
&\left[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\right.\\
&\left.+\,\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\right]\times\\
&\log\left[\frac{k^+\,(x'-w')^2+(\vee-k^+)\,(x'-z)^2}{k^+\,(x'-w')^2+\Lambda\,(x'-z)^2}\right]\times\\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.\\
&\left.-\,f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]
\end{aligned}
\tag{1.10}
$$

$$
\begin{aligned}
\frac{d^3N_{2f\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\log\left[\frac{\vee}{\Lambda+k^+}\right]\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{\left[(x'-w')^2\,(x-w)^2-(x-z)^2\,(x'-z)^2\right]}{(z-w)^2\,(z-w')^2\,\left[(x-z)^2-(x-w)^2\right]\,\left[(x'-z)^2-(x'-w')^2\right]}\\
&\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x')-f^{ac'd'}\,U^{bc'}(z)U^{e'd'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^{d}(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{e}(x)\right]\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-z)^2\,(x-w)^2}{(z-w)^2\,(z-w')^2\,\left[(x-z)^2-(x-w)^2\right]}\,\frac{\left[(x'-z)^2-(x'-w')^2\right]}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\\
&\log\left[\frac{(\vee-k^+)\,(x-z)^2+k^+\,(x-w)^2}{\Lambda\,(x-z)^2+k^+\,(x-w)^2}\right]\times\\
&\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg]\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\left[\frac{1}{\vee}-\frac{1}{k^++\Lambda}\right]\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{1}{(z-w)^2\,(z-w')^2}\\
&\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg]\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')^2\,(x'-z)^2}{(z-w)^2\,(z-w')^2}\\
&\frac{\left[(x-z)^2-(x-w)^2\right]}{\left[(x'-z)^2-(x'-w')^2\right]\,\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\times\log\left[\frac{(\vee-k^+)\,(x'-z)^2+k^+\,(x'-w')^2}{\Lambda\,(x'-z)^2+k^+\,(x'-w')^2}\right]\times\\
&\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg]
\end{aligned}
\tag{1.11}
$$

1.2 Row II

[Done ✓]

Only 4 ρ's

1.3 Row III

[Done ✓]

Only 4 ρ's

1.4 Row IV

[Done ✓]

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} &= \frac{4}{(2\pi)^3}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\int_{y,y'}\Big[\mathbb{A}_{k'}^{\dagger\,(1)\,e'}(p^++k^+,-y')U^{\dagger\,e'c'}(y')U^{d'b}(z)\bar{\mathbb{C}}_{1jik'}^{\dagger\,d'ac'}(p^++k^+,y'-w';p^+,w'-z)\\
&-\bar{\mathbb{A}}_{k'}^{\dagger\,(1)\,c'}(p^++k^+,-y')\bar{\mathbb{C}}_{1jik'}^{\dagger\,d'ac'}(p^++k^+,y'-w';p^+,w'-z)U^{d'b}(z)\Big]\times\Big[\bar{\mathbb{C}}_{1jik}^{dac}(p^++k^+,y-w;p^+,w-z)\\
&U^{bd}(z)U^{ce}(y)\mathbb{A}_k^{(1)\,e}(p^++k^+,-y)-U^{bd}(z)\bar{\mathbb{C}}_{1jik}^{dac}(p^++k^+,y-w;p^+,w-z)\bar{\mathbb{A}}_k^{(1)\,c}(p^++k^+,-y)\Big]
\end{aligned}
$$

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\int_{x,x',y,y'}\frac{(w-z)^m\,(x-y)^k\,(w'-z)^{m'}\,(x'-y')^{k'}}{(w-z)^2\,(x-y)^2\,(w'-z)^2\,(x'-y')^2}\,\frac{p^+\,k^+}{(p^++k^+)^4}\\
&\delta^{(2)}\left[y-w+\frac{p^+}{(p^++k^+)}\,(w-z)\right]\times\delta^{(2)}\left[y'-w'+\frac{p^+}{(p^++k^+)}\,(w'-z)\right]\times\\
&\left[\delta_{km}\,\delta_{ij}-\frac{(p^++k^+)}{p^+}\,\delta_{jm}\,\delta_{ik}-\frac{(p^++k^+)}{k^+}\,\delta_{im}\,\delta_{kj}\right]\times\left[\delta_{k'm'}\,\delta_{ij}-\frac{(p^++k^+)}{p^+}\,\delta_{jm'}\,\delta_{ik'}-\frac{(p^++k^+)}{k^+}\,\delta_{im'}\,\delta_{k'j}\right]\\
&\left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cda}\,U^{ec}(y)\,U^{db}(z)\,\rho^{e}(x)-f^{cda}\,U^{bd}(z)\,U^{ce}(x)\,\rho^{e}(x)\right]
\end{aligned}
\tag{1.12}
$$

Integrating w and w',

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\,\frac{1}{k^+}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_z e^{-ik\,(y'-y)\,\frac{(k^++p^+)}{k^+}}\int_{x,x',y,y'}\frac{(y-z)^m\,(x-y)^k\,(y'-z)^{m'}\,(x'-y')^{k'}}{(y-z)^2\,(x-y)^2\,(y'-z)^2\,(x'-y')^2}$$
$$\left[\frac{\delta_{km}\delta_{k'm'}\,d_\perp\,p^+}{(k^++p^+)^2}+\frac{2}{k^+}\,(\delta_{mk'}\delta_{m'k}-\delta_{m'k'}\delta_{km})+\frac{p^+}{k^{+2}}\,\delta_{mm'}\delta_{kk'}+\frac{1}{p^+}\,\delta_{mm'}\delta_{kk'}\right]\times$$
$$\left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cda}\,U^{ec}(y)\,U^{db}(z)\,\rho^e(x)-f^{cda}\,U^{bd}(z)\,U^{ce}(x)\,\rho^e(x)\right] \tag{1.13}$$

Reducing Wilson lines, we get

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\,\frac{N_c}{k^+}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_z e^{-ik\,(y'-y)}\,e^{-ik\,(y'-y)\frac{p^+}{k^+}}\int_{x,x',y,y'}\frac{(y-z)^m\,(x-y)^k\,(y'-z)^{m'}\,(x'-y')^{k'}}{(y-z)^2\,(x-y)^2\,(y'-z)^2\,(x'-y')^2}$$
$$\left[\frac{\delta_{km}\delta_{k'm'}\,d_\perp\,p^+}{(k^++p^+)^2}+\frac{2}{k^+}\,(\delta_{mk'}\delta_{m'k}-\delta_{m'k'}\delta_{km})+\frac{p^+}{k^{+2}}\,\delta_{mm'}\delta_{kk'}+\frac{1}{p^+}\,\delta_{mm'}\delta_{kk'}\right]\times$$
$$\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x) \tag{1.14}$$

Removed (see the corrected version below). Integrating wrt z and simplifying, we get

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{N_c}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[\frac{k^+}{\vee}\right]\right]\int_{w,w'}e^{-ik\,(y'-y)}\int_{x,y,y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[\frac{1}{\bar\epsilon}-\log\left[4\pi^2\,\mu^2\,(y'-y)^2\right]\right]\times$$
$$\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)$$
$$-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_1^{\frac{k^+}{\vee}}\frac{d\xi}{\xi^2}\int_{w,w'}e^{-ik\,(y'-y)\,\frac{1}{\xi}}\int_{x,y,y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\,\frac{\mathcal{P}_{gg}}{2}\left[\frac{1}{\bar\epsilon}+1-\log\left[4\pi^2\,\mu^2\,(y'-y)^2\right]\right]\times$$
$$\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)$$
$$+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{N_c}{k^+}\int_1^{\frac{k^+}{\vee}}\frac{d\xi}{\xi^2}\int_{w,w'}e^{-ik\,(y'-y)\,\frac{1}{\xi}}\int_{x,y,y'}\left[\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\,d_\perp\,\xi\,(1-\xi)\right.$$
$$\left.+\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[\frac{\xi}{[(1-\xi)]_+}+\frac{(1-\xi)}{\xi}\right]\right]\times\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x) \tag{1.15}$$

Using the fragmentation function

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{N_c}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[\frac{k^+}{\vee}\right]\right]\int_{w,w'}e^{-ik\,(y'-y)}\int_{x,y,y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[\frac{1}{\bar\epsilon}-\log\left[4\pi^2\,\mu^2\,(y'-y)^2\right]\right]\times$$
$$\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)$$
$$+\int_1^{\frac{k^+}{\vee}}\frac{d\xi}{\xi^2}\,D_{g/g}(\xi)\,\frac{d^3N_{\rm LO}}{d^3k}\left(\frac{k}{\xi}\right)$$
$$-\frac{g^2}{4\pi^2}\int_1^{\frac{k^+}{\vee}}\frac{d\xi}{\xi^2}\,\frac{\mathcal{P}_{gg}(\xi)}{2}\,\left[1-\log\left[4\pi^2\,\mu^2\,(y'-y)^2\right]\right]\,\frac{d^3N_{\rm LO}}{d^3k}\left(\frac{k}{\xi}\right)$$
$$+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{N_c}{k^+}\int_1^{\frac{k^+}{\vee}}\frac{d\xi}{\xi^2}\int_{w,w'}e^{-ik\,(y'-y)\,\frac{1}{\xi}}\int_{x,y,y'}\left[\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\,d_\perp\,\xi\,(1-\xi)\right.$$
$$\left.+\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[\frac{\xi}{[(1-\xi)]_+}+\frac{(1-\xi)}{\xi}\right]\right]\times\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x) \tag{1.16}$$

where $D_{g/g}(\xi)=\delta(1-\xi)-\frac{g^2}{8\pi^2}\,\frac{1}{\bar\epsilon}\,\mathcal{P}_{gg}(\xi)+\mathcal{O}(g^4)$ is the Fragmentation function.

Corrected: the z integration. The z integral is ultraviolet finite and infrared (collinear) divergent. In $d_\perp=2-\epsilon$ with the $\overline{\rm MS}$ factor $(\mu^2e^{\gamma_E}/4\pi)^{\epsilon/2}$ (the same scheme as in Group I, Row II),

$$\int_z\frac{(y-z)^m\,(y'-z)^{m'}}{(y-z)^2\,(y'-z)^2}=-\frac{\pi}{2}\,\delta^{mm'}\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\right]-\pi\,\frac{(y'-y)^m\,(y'-y)^{m'}}{(y'-y)^2}+O(\epsilon), \tag{1.17}$$

with trace $-\pi\big[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\big]$ in $d_\perp$ dimensions. Here $\frac{2}{\epsilon}$ is an infrared pole ($\epsilon<0$). The integral is symmetric in $m\leftrightarrow m'$, so the term $\frac{2}{k^+}(\delta_{mk'}\delta_{m'k}-\delta_{m'k'}\delta_{km})$ drops out. With $d_\perp=2-\epsilon$ in the first term ($d_\perp\times\frac{2}{\epsilon}$ leaves a finite -1),

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} = &-\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{\vee-k^+}dp^+\int_{x,x',y,y'}e^{-ik\cdot(y'-y)\frac{p^++k^+}{k^+}}\left\{\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[\Big(\frac{p^+}{(p^++k^+)^2}+\frac{p^+}{k^{+2}}+\frac{1}{p^+}\Big)\right.\right.\\
&\left.\times\Big(\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\Big)-\frac{p^+}{(p^++k^+)^2}\right]+\frac{2p^+}{(p^++k^+)^2}\,\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\left.\right\}\\
&\times\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\rho^{e'}(x')\Big[U^{ce}(y)-U^{ce}(x)\Big]\rho^e(x).\end{aligned}\tag{1.18}$$

In terms of $\xi=\frac{k^+}{p^++k^+}$ (the fraction of the parent carried by the measured gluon), $dp^+=k^+d\xi/\xi^2$ and $k^+\big(\frac{p^+}{(p^++k^+)^2}+\frac{p^+}{k^{+2}}+\frac{1}{p^+}\big)=\xi(1-\xi)+\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}$. The limits are exact: $\Lambda<p^+<\vee-k^+$ is $\frac{k^+}{\vee}<\xi<\frac{k^+}{k^++\Lambda}$.

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} = &-\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{8\pi^4}\,\frac{1}{k^+}\int_{k^+/\vee}^{k^+/(k^++\Lambda)}\frac{d\xi}{\xi^2}\int_{x,x',y,y'}e^{-ik\cdot(y'-y)/\xi}\left\{\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[\Big(\xi(1-\xi)+\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}\Big)\right.\right.\\
&\left.\times\Big(\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\Big)-\xi(1-\xi)\right]+2\xi(1-\xi)\,\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\left.\right\}\\
&\times\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\rho^{e'}(x')\Big[U^{ce}(y)-U^{ce}(x)\Big]\rho^e(x).\end{aligned}\tag{1.19}$$

Corrected: the endpoint $\xi\to1$ (soft unobserved gluon). With $\frac{\xi}{1-\xi}=\frac{1}{1-\xi}-1$ and, exactly, $\int_{\xi_0}^{\xi_1}d\xi\,\frac{g(\xi)}{1-\xi}=\int_{\xi_0}^{\xi_1}d\xi\,\frac{g(\xi)-g(1)}{1-\xi}+g(1)\log\frac{1-\xi_0}{1-\xi_1}$, where

$$\log\frac{1-\xi_0}{1-\xi_1}=\log\frac{(\vee-k^+)(k^++\Lambda)}{\vee\,\Lambda}=\log\Big[\frac{\vee}{\Lambda}\Big]-\log\Big[\frac{\vee}{k^+}\Big]+\log\Big[1-\frac{k^+}{\vee}\Big]+\log\Big[1+\frac{\Lambda}{k^+}\Big],\tag{1.20}$$

the soft endpoint is

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k}\bigg|_{\rm endpoint} = &-\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{8\pi^4}\,\frac{1}{k^+}\left[\log\Big[\frac{\vee}{\Lambda}\Big]-\log\Big[\frac{\vee}{k^+}\Big]+\log\Big[1-\frac{k^+}{\vee}\Big]+\log\Big[1+\frac{\Lambda}{k^+}\Big]\right]\\
&\times\int_{x,x',y,y'}e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\Big(\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\Big)\\
&\times\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\rho^{e'}(x')\Big[U^{ce}(y)-U^{ce}(x)\Big]\rho^e(x),\end{aligned}\tag{1.21}$$

and the rest is the ξ integral above with $\frac{1}{\xi^2(1-\xi)}e^{-ik\cdot(y'-y)/\xi}\to\frac{1}{1-\xi}\big[\frac{1}{\xi^2}e^{-ik\cdot(y'-y)/\xi}-e^{-ik\cdot(y'-y)}\big]$, over the same limits. Your $\log\frac{\vee}{\Lambda}+\log\frac{k^+}{\vee}$ misses $\log(1-\frac{k^+}{\vee})+\log(1+\frac{\Lambda}{k^+})$, and your $\int_1^{k^+/\vee}$ uses $\xi_1=1$ instead of $\frac{k^+}{k^++\Lambda}$.

Corrected: the fragmentation form. With the LO cross section (emitter at y)

$$\frac{d^3N_{\rm LO}}{d^3k}(k^+,\mathbf{k})=\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2k^+}\int_{x,x',y,y'}e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\rho^{e'}(x')\Big[U^{ce}(y)-U^{ce}(x)\Big]\rho^e(x),\tag{1.22}$$

the pole part of this row is

$$\frac{d^3N_{\rm NLO}}{d^3k}\bigg|_{\rm pole}=-\frac{\alpha_s}{2\pi}\,\frac{2}{\epsilon}\int_{k^+/\vee}^{k^+/(k^++\Lambda)}\frac{d\xi}{\xi^3}\,P^{(r)}_{gg}(\xi)\,\frac{d^3N_{\rm LO}}{d^3k}\Big(\frac{k^+}{\xi},\frac{\mathbf{k}}{\xi}\Big),\qquad P^{(r)}_{gg}(\xi)=2N_c\Big[\xi(1-\xi)+\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}\Big].\tag{1.23}$$

The measure is $d\xi/\xi^3$, not $d\xi/\xi^2$: $\frac{d^3N_{\rm LO}}{d^3k}(k^+/\xi,\mathbf{k}/\xi)$ carries ξ/k^+. This is the real ($\xi<1$) part of the collinear counterterm of the gluon fragmentation function. The $\delta(1-\xi)$ part of P_{gg} ($\frac{\beta_0}{2}$ and the soft endpoint) is not produced by this row; it comes with the virtual terms of Group I. So $D_{g/g}\otimes{\rm LO}$ with $D_{g/g}=\delta(1-\xi)+O(g^2)$ cannot be written for this row alone: its $\delta(1-\xi)$ term is the LO cross section itself. In your sign convention ($1/\bar\epsilon$ for the infrared pole), $\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2}{4}=-\big[\frac{1}{\bar\epsilon}-\log\frac{(y'-y)^2\mu^2e^{2\gamma_E}}{4}\big]$: the constant in the logarithm is $e^{2\gamma_E}/4$, not $4\pi^2$.